KEVIN BRYAN | 1692617

THEMATIC ANALYSIS THEMES:

FOR INTRODUCTION

- **The Pedagogical Shift in the GenAI Era**
    - Generative AI is transforming higher education from a focus on banning the technology to integrating it as an active pedagogical partner. The goal is no longer just content delivery, but using AI to foster higher-order critical thinking.
- **"Productivity Paradox" & Cognitive Offloading**
    - Students increasingly use AI as a shortcut to maximize productivity and meet deadlines rather than to learn. This "cognitive offloading" allows them to generate final outputs while bypassing the mental friction required for actual memory retention.
- **Illusion of Truth and Confidence Miscalibration**
    - Because AI generates text with a highly authoritative and fluent tone, students often blindly trust the output without verification. This creates a dangerous "illusion of competence," where a student's confidence in their knowledge is artificially high compared to their actual test scores.
- **Need for "Desirable Friction" (The Proposal Hook)**
    - To combat this false confidence, AI tools must be designed with "speed bumps" that force cognitive engagement rather than just handing out direct answers. This study proposes testing different AI interaction styles to see which type of friction best aligns a student's confidence with their true competence.

FOR SLR

- **Frameworks for Human-AI Collaboration**
    - Existing theoretical models (like the ICAP framework and Mollick's "Seven Ways") map human-AI interactions on a spectrum from passive consumption to interactive dialogue. These frameworks provide the theoretical justification for designing distinct, testable AI personas for this study.
- **The Prerequisite of AI Literacy & Foundational Knowledge**
    - AI is only an effective learning partner if students already possess a baseline understanding of the topic and the literacy to prompt the AI effectively. Without prior foundational knowledge, students cannot evaluate AI hallucinations, turning the tool back into a crutch.
- **The Passive Baseline (The "Crutch" Effect)**
    - When AI is used passively to provide direct answers (such as generating summaries or flashcards), it removes necessary learning friction. Literature shows this boosts immediate performance but ultimately harms long-term retention and skill transfer once the AI is removed.
- **Constructive Friction: The Socratic Tutor**
    - A Socratic AI introduces "constructive friction" by interrogating the student with open-ended questions and refusing to provide direct facts. This pedagogical method forces students to verbalize their reasoning, leading to significantly deeper conceptual understanding.
- **Generative Friction: Reverse Tutoring (The Protégé Effect)**
    - Reverse tutoring creates "generative friction" by having the student explain a concept to an AI acting as a confused novice. This forces students to organize their mental models and quickly exposes gaps in their own understanding, shattering the illusion of competence.
- **Psychological Impacts: Self-Efficacy, Trust, and Technostress**
    - Beyond just test scores, different AI interactions uniquely affect a student's emotional state, trust, and fatigue. Understanding these psychological impacts is crucial, as high-friction methods (like Socratic tutoring) might improve learning but also increase student frustration compared to human teachers.

FOR INTRODUCTION

Theme 1: The Pedagogical Shift in the GenAI Era

Definition (from your notes): Generative AI is transforming higher education from a focus on banning the technology to integrating it as an active pedagogical partner. The goal is no longer just content delivery, but using AI to foster higher-order critical thinking.

Grouped notes:

- **Sallai et al. (2024):** Uni needs to create an environment where AI enhances human intelligence; GenAI should be prevented if it removes opportunities to develop cognitive abilities/social skills. Use of AI is not lazy—educators should see it as an opportunity to boost student interest. Policy: delay adoption of AI in learning new skills by aligning foundation first, then AI becomes the assistant to automate.
- **Wenjuan et al.:** Utilises active theory model—provides a comprehensive lens for examining learning processes (outcomes and facilitation). Why chatbots: lack of personalized learning, lack of realistic learning interaction, need for safe/controllable environment; AI can provide customized feedback and guide in-depth conversations.
- **Integrating GenAI (2024):** Instead of banning AI, focus on decision making & training to enhance human capabilities and future readiness. Guidelines should be flexible to context (faculty/department goals); focus on honesty, ethical use, and citing AI.
- **Mollick & Mollick (2023) – Five Strategies:** AI to provide examples/explanations/analogies, address misconceptions, low-stakes testing, identify knowledge gaps, design materials that reinforce learning. AI as tool so educators can evaluate AI hallucination; help “overwork instructors” by making complex pedagogy accessible.
- **Qian (2025):** Focus towards using GenAI to foster higher-order skills—creativity, critical thinking, learning autonomy, prompt literacy. Future research must focus on exploratory, human-centric models ensuring AI serves as cognitive partner rather than cognitive replacement.
- **Prather et al. (2025):** Trend from banning GenAI towards integrating with hybrid models that combine traditional pedagogy with AI-assisted exploration. Prompt engineering as required competency; GenAI lowering entry barrier for complex computing tasks; human oversight needed.
- **Ma et al. (2025):** GenAI is no longer optional but a necessity; pedagogical shift from memorization to problem solving & active learning (project-based, interdisciplinary). Human-centric competencies (critical judgment, ethical judgement) remain important. Modular, tiered, problem-focused strategy so GenAI serves as catalyst for innovation.
- **Pham (2026):** Critical need for educators to reflect on ethical implications; UNESCO AI Competency Framework as global standards for what students & educators need regarding AI.

- **Alshumaimeri et al. (2026):** Shift in assessment towards the *learning process* (e.g. oral examination, critical AI-output evaluations). Teacher autonomy vital; pedagogical decisions must remain human-centric and ethically grounded.
- **Krause et al. (2025):** GenAI changing how knowledge is accessed and processed. Institutions must move beyond general policies to guidelines addressing unique needs of students & teachers.
- **Xu & Chen (2026):** GenAI as partner in human-centred pedagogy focusing on creative and critical thinking; human as final judge; AI as digital whiteboard; students focus on higher-order skills (synthesis, critique, design), GenAI for scaffolding and automating repetitive tasks.
- **Li (2025):** AI-driven oral assessment for ESL; G-ASR tool; more interactive than teacher in one measure (60% AI, 40% teacher).

Conclusion (Theme 1):
The literature consistently shows a move from prohibition to integration: GenAI is framed as a pedagogical partner that should support higher-order thinking, creativity, and critical evaluation rather than replace them. Success depends on flexible, context-sensitive guidelines, educator training, and assessment that values process and human judgment. The shift is towards human-centric, active learning with AI as a scaffold, not a substitute.

Theme 2: "Productivity Paradox" & Cognitive Offloading

Definition: Students increasingly use AI as a shortcut to maximize productivity and meet deadlines rather than to learn. This "cognitive offloading" allows them to generate final outputs while bypassing the mental friction required for actual memory retention.

Grouped notes:

- **Sallai et al. (2024):** Students find it easier to learn at the start, but mixed perceptions by autumn; overuse → lower quality work, generic essays. **Abbas et al. (cited):** excessive use of ChatGPT reduces academic performance and memory.
- **Wang et al. (2026) – Postgraduate:** Productivity Paradox—AI speeds up drafting but leads to reduction in depth of critical analysis. Requires healthy skepticism and oversight of AI-generated contributions.
- **Hou et al. (2025):** Patterns of reliance: Initial consultation, Verification seeking, **Content delegation** (adopting AI-generated content into final output). Blind reliance → lower cognitive engagement; using AI as critical partner → improved problem-solving strategies.
- **Lodge et al. (2026):** Risk of cognitive outsourcing when students rely heavily on AI to plan, organise, and evaluate; they fail to develop internal self-regulation skills. GenAI not yet supportive of self-regulated learning unless pedagogically engineered to provide hints & scaffold rather than direct answers.

- **Bastani et al. (2025):** GPT Base (direct answer): 48% improvement in practice, but when AI removed, 17% *worse* than no-AI control. AI as crutch—complete task without acquiring underlying skills.
- **Klimova & Pikhart (2025):** Over-reliance on AI can reduce face-to-face interactions, declining interpersonal skills and emotional intelligence → social anxiety, isolation. Blurs line between leisure and academic task.
- **Miller & Thompson (2026):** Students prioritise speed & convenience of AI summaries over rigorous verification of underlying sources.
- **Qian (2025):** AI enhances productivity but risks students outsourcing critical cognitive tasks to the model ("brain drain").

Conclusion (Theme 2):
The Productivity Paradox is well documented: AI improves immediate task completion and perceived ease but often at the cost of depth, retention, and transfer. Reliance patterns (e.g. content delegation, verification-seeking) and lack of scaffolding lead to cognitive offloading and weaker long-term learning. Mitigations include designing for hints and process rather than direct answers, and fostering self-regulation and oversight rather than automation of thinking.

Theme 3: Illusion of Truth and Confidence Miscalibration

Definition: Because AI generates text with a highly authoritative and fluent tone, students often blindly trust the output without verification. This creates a dangerous "illusion of competence," where confidence in knowledge is artificially high compared to actual test scores.

Grouped notes:

- **Sallai et al. (2024):** Convincing tone of AI chatbots gives illusion that output is true. Lack of minimal foundation knowledge makes it hard to validate AI-generated content → curriculum needs to be adjusted.
- **Miller & Thompson (2026):** Students proficient at prompting but lack skills to evaluate synthetic coherence of AI response → overlook hallucination. Eye-tracking and post-task evaluation used to study credibility and trust.
- **Wang et al. (2026) – Postgraduate:** Higher awareness of academic integrity & risk of AI hallucination among postgrads; need for healthy skepticism.
- **Yang et al. (2025):** Students achieve *higher* learning outcomes (test scores) with GenAI than with teachers—but teacher group has *higher self-efficacy*. AI helps performance; teachers are more effective at building student *confidence* in their own abilities. No significant difference in learning satisfaction.
- **Duelen et al. (2024):** Simply flagging content as fake is ineffective; users need to develop internal critical thinking. Socratic method helps users recognise cognitive biases & emotional triggers and become more cognitively active vs passive fact-checking.
- **Pham (2026):** Survey: 100% of sampled students support ChatGPT but are concerned about authenticity & reliability of content.

- **Jivtode (2024):** Data might be wrong → false results; over-reliance can limit critical thinking and long-term development.
- **Lo (2023) – CLEAR:** Reflective component—users should analyse response for accuracy and bias; humans in the loop responsible for final output.

Conclusion (Theme 3):
Fluent, authoritative AI output encourages over-trust and poor calibration between confidence and competence. Evidence shows that performance can rise while self-efficacy and critical verification do not, unless explicitly fostered. Addressing the illusion of truth requires foundation knowledge, reflective evaluation of AI output, and pedagogies (e.g. Socratic) that build internal critical thinking rather than dependence on external fact-checking.

Theme 4: Need for "Desirable Friction" (The Proposal Hook)

Definition: To combat false confidence, AI tools must be designed with "speed bumps" that force cognitive engagement rather than handing out direct answers. This study proposes testing different AI interaction styles to see which type of friction best aligns a student's confidence with their true competence.

Grouped notes:

- **Sallai et al. (2024) – Policy:** Assessment that requires documentation of development to final output (e.g. pitches, video updates); write down steps/strategies to help students draw connections; delay adoption until foundation aligned; encourage students to track and share use of GenAI.
- **Bastani et al. (2025):** Design of AI interaction is more important for learning than model capabilities. GPT Tutor (hints & guidance) → 127% improvement in practice and *no* learning loss when removed; GPT Base (direct answer) → learning loss. GenAI without guardrails can harm learning.
- **Liding et al. (2026):** Compares Targeted vs Socratic hints (coding). Socratic: worse efficiency (more time, attempts, keystrokes), higher error ratio; no evidence Socratic improved long-term outcomes or coding skills. *Counterpoint:* in this domain, targeted approach remained superior despite Socratic being pedagogically sound.
- **Perez et al. (2026):** RAG pipeline gives effective feedback; successfully provides *scaffolding*—guides rather than direct answers; mitigates hallucinations when grounded in course materials.
- **Lodge et al. (2026):** GenAI must be pedagogically engineered to provide hints & scaffold rather than direct answers to support self-regulated learning.
- **Lo (2023) – CLEAR:** Reflective—analyse response for accuracy and bias; Adaptive—keep refining; humans responsible for final output.
- **Hilliger et al. (2026):** Socratic chatbot responds with open-ended questions & reflective prompts rather than direct answers; 85% of educators agreed useful for promoting self-regulated learning.

- **Ding et al. (2024):** Standard LLM gives direct solution and deprives cognitive effort; Socratic back-and-forth encourages verbalising reasoning and correcting misconceptions.
- **SocratiQ (Jabbour et al. 2025):** Engineered to prevent the “direct answer” trap; conversational engine guides through complex topics.
- **Favero et al. (2024):** Bot uses fine-tuning for structured thought-provoking questions; when student is stuck, provides scaffolding → guides to student conclusion; avoids repetition, stays on topic.
- **Jin et al. (2025):** Scaffolding chatbot (follow-up questions, deeper analysis & self-reflection) vs conventional (direct answers). Low-literacy learners struggle with scaffolding bots (frustrating); GenAI should scaffold critical thinking and require foundational AI literacy.

Conclusion (Theme 4):
Desirable friction—scaffolding, hints, open-ended questions, and process-oriented assessment—is widely recommended to align performance with real competence and reduce over-trust. Empirical support is strong in some domains (e.g. math, Bastani et al.) and mixed in others (e.g. coding, Liding et al.), underscoring the need to *test* which interaction styles best improve both learning and confidence calibration in your target context, which is exactly what your proposal aims to do.

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Theme 5: Frameworks for Human–AI Collaboration

Definition: Existing theoretical models (e.g. ICAP, Mollick's "Seven Ways") map human–AI interactions on a spectrum from passive consumption to interactive dialogue. These frameworks provide the theoretical justification for designing distinct, testable AI personas for this study.

Grouped notes:

- **Wenjuan et al.:** Active theory model—comprehensive lens for examining learning processes and facilitation.
- **Mollick & Mollick (2023) – Seven Ways:** AI-Mentor (feedback), AI-Tutor (direct instruction), AI-Coach (metacognition), AI-Teammate (collaborative), AI-Student (learning by teaching), AI-Simulator (practice), AI-Tool (efficiency—warn overuse risks missing core skills). Educators create hard thinking tasks AI can't replicate; teach students to verify AI outputs.
- **ICAP (Chi & Wylie, 2014) – from your methods notes:** Interactive (highest: dialogue, joint construction) > Constructive (generate beyond materials, self-explaining) > Active (physical manipulation, notes, highlighting) > Passive (receive only).
- **Fakour & Imani (2025):** Socratic questioning—thought-provoking questions rather than direct answers; Critical Thinking Dimensions: Analysis, Evaluation, Inference, Self-Regulation. AI to foster intellectual humility and persistent questioning, not replace humans.
- **Debets et al. (2025):** 65% of reviewed chatbots are teaching-oriented; 18% informed by educational theory. Warns: adopting bots for convenience rather than alignment with learning goals; many papers lack theoretical foundations → misguided implementation.
- **Jin et al. (2025):** GenAI literacy essential for productive human–AI collaboration; compares conventional vs scaffolding chatbots; literacy affects ability to steer AI for deeper insights.
- **Kim et al. (2026):** High AI literacy → collaborative approach (planning & evaluation with AI); low literacy → less interactive, linear use. High literacy outperforms on content, structure, expression; takes AI as partner; low literacy over-relies without critical assessment → poor long-term outcomes.
- **Elkhodr & Gide (2025):** GenAI should transform students from informed receivers to critical evaluators; two-phase model: tutorial exercises (refine AI outputs, identify gaps) and assessment tasks (apply to real-world scenarios).

Conclusion (Theme 5):
Frameworks such as ICAP and Mollick's Seven Ways offer a clear spectrum from passive consumption to interactive and constructive engagement. They support the design of distinct AI conditions (e.g. Socratic tutor vs reverse tutoring vs passive tool) and stress that theory-driven design and AI literacy are necessary for effective human–AI collaboration. Your study can explicitly anchor its conditions in these frameworks.

Theme 6: The Prerequisite of AI Literacy & Foundational Knowledge

Definition: AI is only an effective learning partner if students already possess a baseline understanding of the topic and the literacy to prompt and evaluate the AI effectively. Without prior foundational knowledge, students cannot evaluate AI hallucinations, turning the tool back into a crutch.

Grouped notes:

- **Sallai et al. (2024):** Students benefit most from GenAI when they understand the purpose and have basic concepts needed for the task. Lack of minimal foundation makes it hard to validate AI-generated content → curriculum needs adjustment.
- **Kim et al. (2026):** High AI literacy → collaborative, partner-like use; low literacy → linear, over-reliance, poor long-term learning. Instructors should make students flexible cognitively and guide use for high-order reasoning.
- **Jin et al. (2025):** Students with higher GenAI literacy better at steering AI for deeper insights; low literacy struggle with scaffolding bots (frustrating, revert to simple queries). GenAI literacy assessed before experiment (understand, use, evaluate). Need foundational AI literacy to engage with scaffolding systems.
- **Wang et al. (2024) – Catalyst:** Those with stronger critical thinking & positive learning attitude more likely to use AI effectively. Tension between mastering AI and over-reliance; use should be integrated “judiciously”; educators guide prudent use and nurture critical thinking.
- **Ma et al. (2025):** AI literacy in three tiers: Fundamental (logic, ethics), Applied (domain-specific tools), Advanced (create/customise AI workflows).
- **Perez et al. (2026):** RAG grounded in course materials mitigates hallucinations; scaffolding (guides not direct answers) supports learning.
- **Lo (2023) – CLEAR:** Reflective step—analyse for accuracy and bias; humans in the loop responsible for final output. Librarians/teachers teach prompt engineering for better results.
- **Shamsi et al. (2024):** AI as tool for deepening knowledge, not shortcut; fairness, transparency, responsibility; institutions must provide guidelines.

Conclusion (Theme 6):
Across studies, foundational knowledge and AI literacy are prerequisites for beneficial use: they enable verification of outputs, effective prompting, and productive use of scaffolding rather than passive consumption. Curriculum and instruction should build these before or alongside AI use, and your design can control for or measure prior knowledge and literacy when comparing tutoring styles.

Theme 7: The Passive Baseline (The "Crutch" Effect)

Definition: When AI is used passively to provide direct answers (e.g. generating summaries or flashcards), it removes necessary learning friction. Literature shows this boosts immediate performance but harms long-term retention and skill transfer once the AI is removed.

Grouped notes:

- **Bastani et al. (2025):** GPT Base (direct answer): 48% improvement in practice; when AI removed, 17% *worse* than control. GPT Tutor (Socratic) no learning loss. AI as crutch—complete task without acquiring underlying skills.
- **Sallai et al. (2024):** Students that use chatbots too much produce lower quality, generic work; direct outputs bypass mental friction.
- **Liding et al. (2026):** Targeted hints (clear, concise suggestion) superior to Socratic in their coding study—more efficient, fewer repeated errors; Socratic did not improve long-term outcomes here.
- **Soyoof et al. (2026):** ChatGPT feedback one-directional and static (predefined prompts); human feedback outperformed on both immediate & post-test; students trust teacher more; AI feedback mechanical, repetitive, boring → frustrated students, undermined trust.
- **Hou et al. (2025):** Content delegation (adopting AI into final output) and blind reliance → lower cognitive engagement.
- **Lodge et al. (2026):** GenAI not yet supportive of self-regulated learning when it gives direct answers; must provide hints & scaffold instead.
- **Jin et al. (2025):** Conventional chatbot (direct answers) vs scaffolding (follow-up questions); low-literacy learners revert to simple transactional queries with conventional use.
- **Ding et al. (2024):** Standard LLM typically gives direct solution to math problems, depriving students of cognitive effort.
- **Debets et al. (2025):** Primary use shifted from administrative to pedagogical; warning that convenience-driven adoption without learning goals leads to misuse.
- **Jivtode (2024):** Over-reliance limits critical thinking and creativity; long-term can limit cognitive, social, emotional development.

Conclusion (Theme 7):
The passive baseline—direct answers, summarisation, or heavy content delegation—consistently shows short-term gains but learning loss or worse performance when the support is removed. It acts as a crutch that bypasses the effort needed for encoding and transfer. Your study can use a passive/direct-answer condition as the baseline against which Socratic and reverse-tutoring conditions are compared.

Theme 8: Constructive Friction: The Socratic Tutor

Definition: A Socratic AI introduces "constructive friction" by asking open-ended questions and refusing to give direct facts. This forces students to verbalise reasoning and leads to significantly deeper conceptual understanding.

Grouped notes:

- **Tamang et al. (2021):** Socratic method 59% learning gain vs free self-explanation 30% (both for source code comprehension in Java); works regardless of prior knowledge.
- **Duelen et al. (2024):** Socratic AI for disinformation—guiding questions, Island approach; users develop internal critical thinking, recognise biases, more cognitively active than passive fact-checking.
- **GELEX (Yazici et al. 2024):** Prompt engineering + guardrails force LLM to act as Socratic tutor; hybrid expert module enforces pedagogical constraints; reduces workload (expert gives base example, AI builds variations).
- **Bastani et al. (2025):** GPT Tutor (Socratic, hints & guidance): 127% improvement in practice, *no* learning loss when removed (equal to control).
- **Hilliger et al. (2026):** Socratic chatbot: open-ended questions & reflective prompts, no direct answers; 85% educators agreed useful for self-regulated learning; built with GPT-4o-mini; scenarios in CS, language, math.
- **Ding et al. (2024):** Framework to turn LLMs from answer engines into Socratic tutors in math; back-and-forth encourages verbalising reasoning and correcting misconceptions; compare Chain-of-Thought vs Tree-of-Thought for teaching quality.
- **SocratiQ (Jabbour et al. 2025):** Prevents direct-answer trap; conversational engine guides through complex topics; benefits of 1-on-1 Socratic tutoring at scale.
- **Favero et al. (2024):** Fine-tuned bot asks structured thought-provoking questions (Llama-2 7B/13B); when student stuck, scaffolding to student conclusion; good for theoretical concepts; addresses privacy and access.
- **Alshaik et al. (2020):** Socratic method for code—guides student to reason through logic and key concepts; students showed better improvement in programming concepts vs pre-test; breaks code into logical segments requiring cognitive processing.
- **Liding et al. (2026) – counterpoint:** In their coding study, Socratic hints performed worse (efficiency, error ratio) and did not improve long-term outcomes; targeted hints remained superior. Domain and task may matter.

Conclusion (Theme 8):
Socratic tutoring is strongly supported in many contexts (math, conceptual reasoning, disinformation, some programming) for deeper understanding and no learning loss when support is removed. Evidence is mixed in one coding study (Liding et al.), so domain and implementation details matter. Your proposal's use of a Socratic condition is well justified; you can discuss domain (e.g. conceptual vs procedural) and hint design as factors to test.

Theme 9: Generative Friction: Reverse Tutoring (The Protégé Effect)

Definition: Reverse tutoring creates "generative friction" by having the student explain a concept to an AI acting as a confused novice. This forces students to organise their mental models and quickly exposes gaps, shattering the illusion of competence.

Grouped notes:

- **Your design notes (p.19–20):** Reverse Tutoring (WILL USE): Student is teacher; AI acts like a 10-year-old asking clarifying questions ("Wait, but why does X happen").
- **Okita & Schwartz (2013):** Learning by teaching with Recursive Feedback—tutor observes how tutee applies what's been taught. Protégé effect: people learn harder when they know they must teach; significant learning gain after teaching sessions; tutor infers from tutee's outcomes.
- **Observing tutee's mistakes/successes** allows tutor to infer own understanding; if tutee fails, tutor's explanation may have been incomplete → triggers self-correction. Having a third party (tutee) important so knowledge must be robust enough to generalise.
- **Mollick "Seven Ways":** AI-Student (learning by teaching)—student teaches the AI, reinforces own understanding.
- **Kim et al. (2026):** High literacy takes AI as *partner*; collaborative use supports better outcomes—consistent with teaching-to-AI as partnership.

Conclusion (Theme 9):
Reverse tutoring (student teaches AI as naive tutee) is underpinned by the protégé effect and recursive feedback: teaching forces organisation of knowledge and exposes gaps. Few papers in your notes implement it with GenAI directly, so your study can fill a gap by comparing it to Socratic and passive conditions and testing whether it improves both learning and confidence calibration.

Theme 10: Psychological Impacts: Self-Efficacy, Trust, and Technostress

Definition: Beyond test scores, different AI interactions affect students' emotional state, trust, and fatigue. Understanding these impacts is crucial, as high-friction methods (e.g. Socratic) might improve learning but also increase frustration compared to human teachers.

Grouped notes:

- **Yang et al. (2025):** Teacher group had *higher self-efficacy* than AI group; AI group had higher test scores. AI good for task performance; teachers better at building *confidence* in own abilities. No significant difference in learning satisfaction (both satisfied).
- **Soyoof et al. (2026):** Human feedback outperformed ChatGPT (immediate & post-test). Trust: students trust teacher more; AI feedback mechanical, repetitive, boring → frustration, undermined trust. Feedback must be collaborative and within ZPD; teachers better at identifying and staying within ZPD than generic AI (Vygotsky).

- **Klimova & Pikhart (2025):** AI has dual impact on wellbeing—supportive but also stressor (fatigue, technostress). Over-reliance → less face-to-face → decline in interpersonal skills, emotional intelligence → social anxiety, isolation. Blurs leisure vs academic.
- **Wang et al. (2024) – Catalyst:** Tension between mastering AI and over-reliance (undermining independent thought); risk of decline in cognitive skills and academic integrity; use "judiciously"; nurture critical thinking.
- **Shamsi et al. (2024):** Ethical use; fairness, transparency, responsibility; guidelines for institutions; tool for deepening knowledge.
- **Abbes et al. (2024):** Gamification + GenAI for personalized learning; adjust to user personality, values, motivation; Education 5.0—self-assessment, ownership of learning; field still lacks models that dynamically adapt content and engagement.
- **Gunawardana et al. (2025):** GenAI for ADHD-friendly LMS interfaces; reduces cognitive noise; minimalism, focus states, standardised palettes; more effective and empathetic design; revolutionary for adaptive personalised learning.
- **Hilliger et al. (2026):** 85% agreed Socratic chatbot useful for self-regulated learning and easy to learn and use—positive usability and perceived usefulness.
- **Jin et al. (2025):** Low-literacy learners find scaffolding bots *frustrating* and revert to simple queries—psychological cost of high-friction design for some learners.

Conclusion (Theme 10):
Psychological outcomes are not redundant with performance: AI can raise scores while self-efficacy stays lower than with a human teacher, and trust and satisfaction depend on feedback quality and interaction design. High-friction methods (Socratic, scaffolding) can improve learning but may increase frustration for low-literacy or time-pressed students. Your study can measure self-efficacy, trust, and perceived difficulty/fatigue alongside knowledge and confidence calibration to capture these trade-offs.

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My Research Question:

1. Investigating the Impact of Generative AI Tutoring Styles on Student Knowledge Acquisition and Confidence Calibration

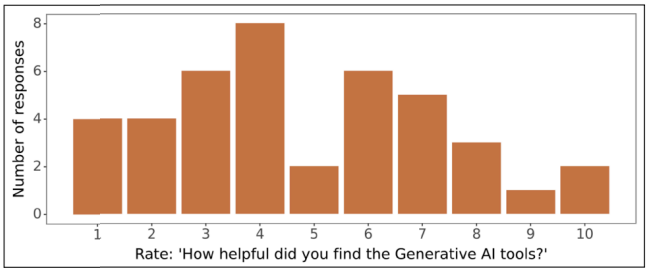
Literature Findings:

1. Approach Generative AI Tools Proactively or Risk Bypassing the Learning Process in Higher Education (Sallai et al., 2024)

Cons:
- Raises the concern of “death” of essays, which has risen from COVID because of increase in online take-home exams → students cheating by utilizing AI to write + paraphrase against plagiarism tools.
- Unreliable, inaccuracies in these tools
- Convincing tone of AI chatbots give illusion that output is true

Pros:
- GenAI can help non-native English speakers linguistically serve as teaching assistant
- GenAI minimizes pressure of homeworks/exams

Keynotes:
- Paper is about undergrad/postgrad students and how they use AI
    - ChatGPT used by almost all undergraduate & postgrad
        - Followed by Grammarly, then Co-pilot
    - They claim that it made it easier for them to learn at the start, but on the autumn term students have mixed perceptions



    -
    - Students in management find the tools less beneficial compared to quantitative courses (data sciences)
    - If Subject pace/delivery is easy to follow they less likely depend on AI
- Primary Analysis:

        - Students benefit most from GenAI when they understand the purpose and have basic concepts needed to complete the task.
            - AI is to help with the *meaningless* task
- Uni needs to create an environment where AI enhances human intelligence
- GenAI should be prevented if it removes opportunities to develop cognitive abilities/social skills
- Students that uses chatbots too much produce lower quality work → essays are generic
- Use of AI is not lazy → educators should realize this as an opportunity to boost student's interest
- Lack of minimal foundation knowledge makes it hard to validate Ai generated content → curriculum needs to be adjusted
- Abbas et al: excessive use of chatgpt reduces academic performance + memory
- Policy recommendation from the paper:
    - Assessment that requires documentation of the development to the final output, example: pitches/video updates.
    - Write down steps / strategies for students to help students draw connections
    - Delay adoption of Ai in learning new skill by aligning the foundation first, then AI becomes the assistant to help automate
    - Encourage student to track and share use of GenAI to support individual learning

2. The who, why, and how of ai-based chatbots for learning and teaching in higher education: A systematic review (Wenjuan et al.)

Utilizes active theory model → provides a comprehensive lens for examining learning processes (both outcomes but also facilitation of learning).
Wh?:
- undergraduates are more inclined to use AI chatbots

Why?
- Lack of **personalized learning experience**
- Lack of realism learning **interaction**
- Limitation of content/presentation
- Safe and controllable learning environment
    - Provide customized feedback, **flexibly adapt to student's needs**
    - Able to guide students to participate in **in-depth convos**

3. Integrating Generative AI in University Teaching and Learning: A Model for Balanced Guidelines" (2024)

Pros:
- AI provides instant feedback + adaptive learning styles
- ChatGPT increases student engagement + satisfaction

Conts:
- Over-reliance & Procrastination, memory loss, decline in learning competence
- Plagiarizing in assignments

Proposed Balanced Approach:
- Instead of banning AI, Focus on decision making & training to enhance human capabilities → future readiness to enter Ai-driven industry
- Guidelines should be flexible to the context (faculty/departments' goal)
- Focus on honesty and ethical points, and fair use of AI such as citing if its from AI

Using AI to Implement Effective Teaching Strategies in Classrooms: Five Strategies, Including Prompts (Mollick & Mollick, 2023)

Strategies for educators on how AI can be effective:
- Provide example + explanations, analogies for abstract concepts
- Address student misconceptions → identify common misunderstandings
- Low Stake Testing → automate quizzes/tests creation
- Identify knowledge gap by assessing where students are struggling
- Help design materials that reinforce learning over time

Cautions
- Instructor/educate must use Ai as a tool, so it can evaluate AI hallucination
- Same prompt can product different result, try multiple times as needed
- It is to help "overwork instructors" by making complex pedagogy accessible

4. Assigning AI: Seven Ways of Using AI in Class (Mollick & Mollick, 2023)

- AI-Mentor (Providing Feedback): Acts as a guide to give students immediate, constructive feedback on their work.
- AI-Tutor (Direct Instruction): Provides personalized, step-by-step tutoring to help students master new concepts.
- AI-Coach (Increasing Metacognition): Prompts students to reflect on their learning process and think through problems.
- AI-Teammate (Collaborative Intelligence): Functions as a partner in group work, helping to brainstorm or provide diverse perspectives.
- AI-Student (Learning by Teaching): The student teaches the AI, which helps reinforce the student's own understanding.

- AI-Simulator (Practice Opportunities): Creates scenarios for students to practice skills in a safe, simulated environment.
- AI-Tool (Efficiency): Used for tasks like summarizing or outlining, though the authors warn this risk-averting role can lead to students missing out on developing core skills if overused.

Students must also be taught to verify AI outputs while educators. Educators also should create hard thinking tasks that AI can't replicate.

5. The CLEAR path: A framework for enhancing information literacy through prompt engineering (Lo, 2023)

CLEAR framework: a pedagogical tool designed to help users from lazy prompting to prompt engineering

- C: Concise, use direct instructions to keep Ai focused on core task
- L: Logical, information presented in step by step/naturally, AI will be more coherent
- E: Explicit, Tell the output you would look like (length, format, tone, points to include)
- A: Adaptive, if answer is general, pivot because prompting is iterative, keep refining
- R: reflective, users should analyze the response for accuracy and bias

Temparature:

Low 0.2 → focus on factual/boring answer

High 0.9 → creativity & word choices but increase risk of hallucinations

The paper argues that librarians should lead teachers to teach students how to engineer prompts to get better search results; however, humans in the loop are important because humans are responsible for the final output.

6. Exploring the effects of artificial intelligence on student and academic well-being in higher education (Klimova & Pikhart, 2025)

Ai has a dual impact on well being, it can be supportive but can be stressor as it can trigger fatigue or technostress.

Over-reliance on AI can also cause the student to reduce face2face interactions, declining the interpersonal skills and emotional intelligence → leading to social anxiety, isolation

It also blurs the line between leisure activities & academic task as it has become indispensable

7. Catalyst for future education: An empirical study on the Impact of artificial intelligence generated content on college students' innovation ability and autonomous learning (Wang et al., 2024)

Uses SEM (structural equation modeling) and meditation model, analyzes relationship between critical thinking, learning attitude, and the use of AI-generative content, innovation entrepreneurship abilities, and autonomous learning abilities.
Findings:

- Those with stronger critical thinking & positive learning attitude are more likely to use AI effectively
- Ai fosters both innovation & entrepreneurship abilities & autonomous learning
- The risk is in the decline of cognitive skills & concern regarding academic integrity.
    - Author argues that there is tension between student mastering AI & risk of over reliance (undermining independent thought)
- A central argument is use of AI should be integrated "judiciously" within the curriculum
- Educators should guide students in using AI prudently, discouraging over reliance and nurture critical thinking

8. Impact of Artificial Intelligence (AI) in Education on Students' Academic Development: Present and Future Prospects (Jivtode, 2024)

Ai has transformed teaching, improved student engagement by providing information based approach
Beyond pedagogy, Ai reduces administrative burden on educators, allowing more interaction with students.
Ai enables intelligent tutoring system that adapts to the student needs in real time, enhancing student engagement and making learning more immersive
Limitations of AI in education:

- Lack of human touch
- Data might be wrong → leading to false results
- Limits critical thinking and creativity development → over reliance → in the long term can limit students development (cognitive, social, emotional)
- Privacy concerns, ethical & bias issues
- AI can't guide like what teachers can do

Impacts of AI Usage Ethically on Students (Shamsi et al., 2024)

Adaptive learning systems improve academic results, however there are technical & ethical hurdles especially in contextual understandings in current Ai tools.

- AI must adhere to principle of fairness, transparency and responsibility
- Institution must provide guidelines to govern ethical use
- It should be a tool for deepening knowledge not shortcut.

Chatbots in education: A systematic review of objectives, underlying technology and theory, evaluation criteria, and impacts (Debets et al. 2025)

Primary use of chatbots in recent years has shifted from administrative assistance to pedagogical support,

- 65% of the reviewed chatbots are teaching oriented that focus on enhancing student knowledge & provide feedback
- 18% (20/71 papers) informed their chatbot is integrated with educational theory
- Author warns that adopting bots just because they are convenient not because they are aligned with learning goals
- Many papers lack theoretical explanations & foundations for the chatbot which may lead to misguided implementation

Integrating Generative AI in Cybersecurity Education: Case Study Insights on Pedagogical Strategies, Critical Thinking, and Responsible AI Use (Elkhodr & Gide, 2025)

- Argues that GenAI should transform students from informed receivers to critical evaluators → require students to critique AI → fostering higer order thinking skills rather than memorization
- Implemented a pedagogical model that consist of 2 distinct phases
    - Tutorial exercises → students refine AI outputs and identify the gaps in the AI's logic
    - Assesment Tasks → apply ai generate outputs to real world scenarios
- GenAI enhanced students ability to bridge theoretical knowledge to practical application, allowes generate risk assessments, enabling students to focus their cognitive effort on high level refinement
- AI must be use as instrucitonal scaffold supporting studnet, they should be thought the fundamental pedagogical strategy to verify AI's output
- GenAI most effective when embedded in real world scenarios, static AI use is discouraged, studnet must refine AI assisted outputs

To Tell or to Ask? Comparing the Effects of Targeted vs. Socratic AI Hints (Liding et al., 2026)

Compares Targeted hints (clear, concise suggestion) with Socratic hints (questions, analogies, examples). In the short term, Socratic performs worse in terms of efficiency, requiring more time, attempts and keystrokes. It also leads to higher error ratio (repeated errors 5%, and 14.7 in long term), they found no evidence that socratic hints improve long term outcomes or coding skills. Although having a higher level of difficulty, it reduces debugging efficiency. Despite the hints from Socratic being pedagogically sound, the targeted approach remained superior.

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A Comparative Study of Free Self-Explanations and Socratic Tutoring Explanations for Source Code Comprehension (Tamang et al., 2021),

Compares 2 instructional strategy in understanding programming code, free-elf explanation & socratic in learning java programming

- Socratic method achieved 59% learning gain. While the free self-explanation achieve 30% learning gain, outperforming by 29%
- Both works well regardless of prior knowledge (no significant difference)
- All 105 participant has prior knowledge of programming
    - Briefed about the goal
    - Given 6 source code example called **main task**,
    - They take post test about content in the pre-test and main taks for final score

From teachers to chatbots: Scaffolded corrective feedback and student trust in online L2 English classrooms (Soyoof et al., 2026)

investigates how scaffolded corrective feedback (SCF) provided by a human teacher compares to that provided by ChatGPT in terms of learning outcomes and student trust

- Student with human feedback outperform chatgpt both immediate & post test
- Qualitative Analysis shows Teacher has advantage in trust, student are able to trust teacher more than AI
- Ai feedback are found to be mechanical, repetitive, and boring, and generation sto[[ong frustrated student & undermined their trust
- ChatGPT feedback are one directional and static due to predefined prompts
- AI should be used by teachers in the loop to make it meaningful
- Grounded in Vygotsky's sociocultural theory, the paper argues that for feedback to be effective, it must be collaborative and situated within the learner's **Zone of Proximal Development (ZPD)**. Teachers were better at identifying and staying within this zone than the generic AI.

The test is conducted with 40 learners of EnglishForeignLanguage, test is for accurate use of english article (a, an, the) → one of most frequent & persistent errors made by L2,

- Pre test given, 2 post test (1 week after (short term effect), and 1 month after (durability of knowledge))
- 20 fill in the blank items where students had to supply the correct article
- Thematic analysis with semi structured interviews

GELEX: Generative AI-Hybrid System for Example-Based Learning (Yazici et al., 2024)

Novel system combining LLM procedural with constraints to improve student learning from examples.
Example based learning are usually static, can't adapt to students misconception. GELEX uses hybrid architecture where LLM generates content, but the an "expert module" efncores pedagogical constraints to ensure the examples are logical

- Use of Prompt engineer for pedagogy with **guardrails can force LLM to act as socratic tutor**
- Reduces workload, because expert can just give base example and AI can build dozens of variation based on students needs
- Tested for logic and procedural tasks

Putting Things into Context: Generative AI-Enabled Context Personalization for Vocabulary Learning Improves Learning Motivation (Leong et al., 2024)

Explores how LLM can be use to personalize contextual sentences into vocabulary learning to match students individual interests

- Personalizing learning context can improve motivation and engagement compared to using generic non personalized examples
- Can incorporate vocabulary into a chosen interest such as cooking/scifi/sports
- Students are more likely to recall and apply new words when the word is used in topics that have interest in
- Make sure the "interesting" context does not distract the learner
- Compares personalized based on user profile, and generic non personalized, uses diverse group of learners to improve their english proficiency
- Measurements: Motivation (IMMS), Learning Outcome (Post test using word definition recall and usage), Qualitative feedback with semi-structured interviews on naturalness and helpfulness of the AI-generated sentences

The authors conclude that LLM-enabled personalization is a transformative tool for language education, offering a way to make traditionally "dry" subjects like vocabulary building more immersive and personally relevant.

Generative AI Based on Deep Knowledge Tracing for Academic Insight and Intelligent Learning Strategy Optimization (Wang, 2026)

Explores integration of deep learning models with gen AI to create prices & adaptive education environments.
→ model can track students degree of mastery in real time, allows educators to identify specific knowledge gaps
→ optimizes student learning strategies, AI can recommend resources or paths tailored to what the student hasn't mastered yet.

Socratic AI Against Disinformation: Improving Critical Thinking to Recognize Disinformation Using Socratic AI (Duelen et al., 2024)

Development of an AI driven tool designed to bolster resistance to fak news by using guiding questions:

- Simply flagging content as fake is ineffective, users need to develop internal critical thinking to evaluate themselves
- Uses a series of structured questions based on the ***Island approach*** to guide user through a critical evaluation of th source
    - Island approach: a qualitative projective technique to evaluate large number of attributes, features, or scenarios
- **Socratic method helps users recognize their own cognitive biases & emotional triggers**. It also empower users to be more cognitively active compared to passive consumption of fact checking

Enhancing Adaptive Personalized Learning Interfaces with Generative AI for Individuals with ADHD (Gunawardana et al., 2025)

Gen Ai to be used as ADHD-friendly user interfaces for LMS

- Traditional LMS contains cognitive noise (distraction, complex layouts, vague instruction)
- Gen AI can act as accessibility architect by using specific accessibility focused personas and constraints in prompts making it adhd friendly
    - Minimalism, explicit focus state and standardized colro palettes
    - Students can use genAI to adjust the interface according to their needs
    - Leads to a more effective and empathetic interface design
- Gen AI is revolutionary for adaptive personalized learning

Generative AI and Gamification for Personalized Learning: Literature Review and Future Challenges (Abbes et al., 2024)

Examines genAI & gamification to create highly adaptive & engaging educational environment

- Author argues that gamification adjusted to user personality, value & motivation can make the learning process mor fun and rigorous
- GAI can be the cornerstone of **education 5.0**, prioritizes personalized and adaptive learning experience over standardized instruction
    - Education 4.0 -> alignment of education practices with 4th industrial revolution, use of big data, iOT, and AI to support learning preparing students for highly digitalized workforce
    - Education 5.0 -> Human-centric personalization, individualized & adaptive based on specific personality, values and motivations. Encourages self assessment and empower students to take ownership of their own learning journey using AI driven tools.
- Practical examples can be seen in Quizizz AI (Wayground) for self assessment allowing student tot take ownership of their learning journey

- However currently the field lacks models that are able to dynamically adapt the content generate & engagement strategy to ensure AI driven environments are motivation and effective.

Generative AI without guardrails can harm learning: Evidence from high school mathematics (Bastani et al., 2025)

- Explores design of AI tutoring tools impact on performance in math
- Shows boost in practice, but leads to deficit when tasked to perform tasks with no AI assistance
- Uses GPT Bases → direct answer, and GPT Tutor → socratic tutor with hints & guidance
    - Base → 48% improvement in practice, Tutor 127% improvement in practice
    - When Ai access removed, GPT base perform 17% worse than no AI control group
    - GPT Tutor did not show learning loss, equal to control group
    - AI can act as a crutch, allows student to complete task without acquiring underlying skills

Study structure:

- 955 students -> 3 groups → base, tutor, control,
- 3 phase → practice (using tools), in-tool testing (measure performance while tools active), transfer phase: student tested without AI with similar problems
- Uses Mathematics topics, ensuring the results were relevant to standard education
- GenAI can be a double edged sword in education, can improve productivity & immediate performance, but lacks guardrails and risks turning into answer machines. Emphasize that design of AI interaction is more important for learning than the model capabilities itself.

Integrating Generative AI with Human-Centered Pedagogy: An Innovative Path for Vocational Education" (Xu & Chen, 2026)

Integration of genAI into vocational training,

- Author argues that genAI shouldn't just be a tool for content delivery but a partner in human centered pedagogy that focuses ont eh students creative and critical thinking
- Case based evidence where AI was successfully integrated:
    - Art & design → image generation
    - Computer Science → RAG that provides context ware answers from notes
    - Math → automated analysis reports for teacher to gain insight
- Human must remain the final judge of quality and relevance, Ai to be used as digital whiteboard
- GenAI allows students to focus on higher order skills such as synthesis, critique, innovative design, and have genAI provide creative scaffolding and automating repetitive tasks

Students-Generative AI interaction patterns and its impact on academic writing" (Kim et al., 2026)

Students with high AI literacy adopted a collaborative approach, engaging AI in planning & evaluation, while low AI literacy is less interactive choosing to ideate & evaluate independently and uses AI in a linear fashion
Student with high literacy outperform across all evaluation rubric, content structure/organization & expression
High literacy takes AI as a partner rather than a mere tool, and low literacy over-rely on AI generated content without critical assessment → poor long term learning outcome.
Instructors should focus on making students flexible cognitively, and guide them to use AI for high order reasoning rather than just tools for content generation.

Evolution of Learning: Assessing the Transformative Impact of Generative AI on Higher Education" (Krause et al., 2025)

Students primarily utilize GenAI for practical, task-oriented activities like assignment writing and exam prep (creating summaries)
GenAi isn't just a tool but an emerging tech that is changing how knowledge is accessed and processed in academic settings.
Institutions must move beyond general policies to develop guidelines that address the unique needs faced by both students & teachers.

The impact of Generative AI on students' learning: a study of learning satisfaction, self-efficacy and learning outcomes" (Yang et al., 2025)

Compare effectiveness of gen ai assisted learning compared to traditional teacher for college students. Students achieve higher learning outcomes when (test scores) when assistant in gen AI compared to teachers.
Students in the teacher group has higher self-efficacy than the AI group, → AI helps students perform better, but teachers are effective in building student confidence in their own abilities.
No significant difference in learning satisfaction, both are satisfied,
AI can be powerful for enhancing task performance & completion, but it can't replicate psychological or motivational support provided by human instructors.
Metrics measured: Learning satisfaction, self-efficacy, and learning outcome using a post-test

Supporting Teaching and Learning Computational Thinking Skills with Generative AI in Computing Education" (Benedetti, 2025)

CT is important skill for achieving goal in computation driven world, beyond just how to code, GenAI can support CT dimensions such as abstraction, decomposition, algorithmic thinking & pattern recognition, Gen Ai is powerful ally in teaching computational thinking, focusing on the process of thinking & problem solving rather than just the final code output, allows student to develop robust CT skill that can be use for various scientific & professional domains.

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Pedagogical Applications of Generative AI in Higher Education: A Systematic Review of the Field (Qian, 2025)

How higher education adapted to genAI during 2 years
→ focus towards using genAI to forst higher-order skills → creativity, critical thinking, learning autonomy & prompt literacy
→ AI enhances productivity, but risk students outsourcing critical cognitive tasks to the model (brain drain)
→ prompting has become a foundational academic competency, essential for students to effectively steer AI towards a more educational outcome
→ future research and practice must focus on exploratory, human-centric models ensuring AI serves as a cognitive partner rather than cognitive replacement.

Bridging pedagogy and technology: a generative AI and IoT approach to transformative English language education (Li, 2025)
→ explores development of AI-driven oral assessment tool for english as a second language education
→ G-ASR tool precision accuracy of 94,7% on native speaker, 86.6% on non-native speaker
→ more interactive than teacher, Ai 60%, teacher 40%

Chatting with a Learning Analytics Dashboard: The Role of Generative AI Literacy on Learner Interaction with Conventional and Scaffolding Chatbots" (Jin et al., 2025)

GenAI literacy effect with different types of Ai chatbots integrated into learning analytics dashboard
→ students with higher genAI literacy are better at steering the AI to provide deeper insights into their own learning pattern.
→ compares 2 Ai chatbots:

- Conventional → direct answers
- Scaffolding → guide using follow up questions → asking deeper analysis & self-reflection

Learners with low literacy struggle with scaffolding bots, finding it frustrating and often reverting to simple transactional queries
GenAi can become an interpreter between the raw data to student's learning goal
Study Strategy:

- Participants were assessed on their GenAI literacy → ability to understand, use and evaluate genAi models before experiment
- Randomized assignment of conventional/scaffolding
- Analysis from: Chatlogs, Dashboard interactions, self-reported surveys

GenAI is essential for productive human-AI collaboration in learning, but it shouldn't just provide answers, it should scaffold students'own critical thinking process, provide the student the foundational AI literacy to engage with the system.

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The Impact of Generative AI on Postgraduate Study (Wang et al., 2026)

How postgraduate utilize genAI and the influence on their research and academic tasks

- Postgrad use genAi for high level research support → literature synthesis & data interpretation
- Productivity Paradox, Ai speeds up drafting but lead to reduction in the depth of critical analysis
- Higher awareness of academic integrity & risk of AI hallucination
- Study strategy uses AI usage log → documenting instances they interacted with the AI, then semi-structured interviews to explore the changing relationship
  Requires humans to have a healthy skepticism or oversight of all the AI-generated contributions.

Navigating the New Information Landscape: A Study of Generative AI-Powered Search Engines in Higher Education (Miller & Thompson, 2026)
→ AI driven search tools are transforming student research habits
→ moving away from list based search result to AI summaries that synthesizes information from multiple sources
→ Student prioritizes speed & convenience of AI summaries over rigorous verification of underlying sources
→ Students are proficient at prompting, but lack the skills to evaluate synthetic coherence of Ai response → overlooking hallucination
Study Strategy:
→ 85 students are tasked to research on a multi-sided social-technical issue (ethics of deep sea-mining), splitted into 2 groups of traditional search engine and AI powered, uses Eye tracking to see which part of AI summaries students focused on, post task eval to rate the credibility of claims and trust, reflection interviews to see how their mental model works

Generative AI to support self-regulated learning: a systematic review of the literature (Lodge et al., 2026)

→ same as previous paper, the risk of cognitive outsourcing, when student rely heavily on AI to plan organize and evaluate their work, student dail to develop internal self regulation skills
→ GenAi arent yet supportive of self regulated learning, they must be pedagogically engineered to provide hints & scaffold rather than direct answers

Beyond the Hype: A Comprehensive Review of Current Trends in Generative AI Research, Teaching Practices, and Tools (Prather et al., 2025)

→ Ai also shifting the skills required by software dev, from syntax heavy coding to higher level architectural design & AI augmented problem solving
→ there is a trend from banning genAI towards integrating it into teaching practices by using hybrid models that combine traditional pedagogy with Ai assisted exploration

→ prompt engineering is not just a technical trick but a required competency in future computing professionals
→ GenAI are Lowering the entry barrier for complex computing tasks, student can engage with advanced concepts
→ AI can generate plausible outputs, therefore human oversight is needed

Large language models for education: An open-source paradigm for automated Q&A in the graduate classroom (Perez et al., 2026)

→ Ai driven pipeline gave effective feedback same as teaching assistants in terms of accuracy & specificity
→ the effectiveness of the RAG in mitigating hallucinations to ensure it is grounded just by the courses provided materials
→ this open source paradigm can provide high quality education and overcome cost & accessibility barrier
→ successfully provide scaffolding → give guides than just direct answers
→ responses from AI compared against human experts through blinded review where evaluator doesn't know the source of the response.

Measuring undergraduate students' reliance on Generative AI during problem-solving: Scale development and validation (Hou et al., 2025)

Study specifies patterns of reliance such as:
→ Initial consultation → using Ai to kickstart brainstorming or decomposition of problem
→ Verification seeking → dependent on AI to check the logic/accuracy of their the student's generated solution
→ Content delegation → adopting Ai generated content into their final output
Student who is blindly reliance on AI experience lower cognitive engagement, while those who uses Ai as critical partner showed improved problem solving strategies

Retrieval Practice Consistently Benefits Student Learning: a Systematic Review of Applied Research in Schools and Classrooms" (Agarwal et al., 2021)

Retrieval learning is where students recall/recognize info from their previously learned in order to strengthen memory and comprehension.
Retrieval practice is one of the most reliable strategy → out of 50 experiments, 57% revealed medium to large benefits of student learning
It is powerful but can be enhanced by providing feedback after retrieval attempt to enhance learning gains & correct misconceptions.

Preparing Students for an AI-Driven World: Generative AI and Curriculum Reform in Higher Education (Ma et al., 2025)

Argues that GenAI is no longer optional curriculum but a necessity in higher education survival due to the rapid professional landscape.
Ai literacy can be put into 3 tier:
→ Fundamental concept: Understand basic logic and ethics of AI
→ Applied use: mastering domain-specific AI tools to enhance productivity
→ Advanced technique: develop the ability to create or customize AI workflow

- The major finding of the paper is the pedagogical shift from memorization to problem solving & active learning strategies, such as project-based or interdisciplinary collaboration.
- Human-centric competencies also remain important such as critical judgment, ethical judgement

By adopting a strategy that is modular, tiered, and problem-focused, institutions can ensure GenAI serves as a catalyst for innovation rather than academic integrity thread.

Scaffolding Learning Scenarios with a Socratic Chatbot: Insights from Educators (Hilliger et al., 2026)

Investigates adoption of instructional design of Socratic chatbot to support student self reflection
Socrative chatbot serves 2 roles:

- Provide personalized support for students
- Assist teachers with instructional design to rethink teaching methods

85% of educators agreed that the AI Chatbot is useful for promoting self-regulated learning and is easy to learn and use.
The socratic chatbot responds with open-ended questions & reflective prompts rather than providing direct answers, which fosters critical thinking.
Built using Gpt-4o-mini, conducted with 44 higher education researchers & practitioners, given a demo of the tool & then tasked with proposing their own learning scenarios.
Collected data from 20 participants through online survey with 5 likert scale to measure usability and usefulness.
8 learning scenarios primarily in computer science, language learning, mathematics, which were analyzed to understand how educators envision integrating the bot into their workflows.

Boosting Large Language Models with Socratic Method for Conversational Mathematics Teaching (Ding et al., 2024)

Framework to transform LLMs into answer engines for Socratic tutors, in mathematics education.

- Identified that standard LLM typically provide direct solution to math problems,which deprive students of the cognitive effort
- The back & forth dialogue of the socratic chatbot encourages students to verbalize their reasoning process, identifying and correcting underlying misconceptions.

- Compare chain of thought and tree of thought to see which is better, models are not just evaluated on the correctness of the final answer but also the teaching quality.
- The methodology involved analyzing long-context dialogues, interaction turns without losing the context of the student's specific struggle.

SocratiQ: A Generative AI-Powered Learning Companion for Personalized Education and Broader Accessibility (Jabbour et al., 2025)

Socratiq → provide benefits of 1on1 socratic tutoring, which is traditionally resource-intensive human process to students at scale through genAI
The platform is engineered to prevent the "direct answer" trap, uses a conversational engine that guides students through the complex topics.

Teaching and Learning in the Age of Generative AI: Evidence-Based Approaches to Pedagogy, Ethics, and Beyond (Pham, 2026)

Emphasizes the critical need for educators to reflect on ethical implications of using ChatGPT, dall-e and gemini in educational settings.
Global standards to define competencies required for the AI era:

- UNESCO AI Competency Framework a structured approach to what students & educators need to know do and understand regarding AI

Transforming Education: The Impact of Generative AI

While GenAI enables increased productivity, it poses a dilemma due tot he lack of established rules & moral standards in education
A survey shows that 100% of sampled students support ChatGPT, but they are concerned on the authenticity & reliability of the content generated.

The Use of Generative AI in Education: Opportunities, Challenges, and Future Directions" (Alshumaimeri et al., 2026)

The ability of Ai to generate high quality essays and reports necessitates a shift in assessment towards the LEARNING PROCESS, such as incorporating oral examination and critical AI-output evaluations.
Highlights the daunting challenges such as algorithmic bias, data privacy concern and also potential depersonalization in the teach-student relationship.
AI can assist,- but teacher autonomy is vital to be preserved to ensure that pedagogical decision remain human-centric and ethically grounded.

“Are You Really Sure?” Understanding the Effects of Human Self-Confidence Calibration in AI-Assisted Decision Making (Ma et al., 2024)

The authors argue that appropriate reliance on AI is not just about understanding AI's limitations, but also about calibrating one's own self-confidence. They suggest that many instances of "inappropriate reliance" (over-reliance or under-reliance) stem from a mismatch between a human's confidence and their actual correctness, rather than just a misunderstanding of the AI.

- **Mismatch Leads to Error**: Discrepancies between human confidence and actual correctness significantly increase error rates in final decisions.
- Correlation with Reliance: A participant's Expected Calibration Error (ECE) positively correlates with both Under-reliance and Over-reliance.
- **AI Confidence is Not Enough**: Simply showing the AI's confidence levels does not automatically improve a human's self-confidence calibration or overall task performance
- Think the Opposite (Think): Effective at improving self-confidence calibration and task performance, but at a high cost to user experience (perceived as complex and mentally demanding).
- **Thinking in Bets (Bet):** Improved task performance and showed a trend in reducing over-confidence, but was ineffective at improving overall self-confidence calibration.
- **Calibration Status Feedback (Feedback):** Highly effective at improving confidence calibration and task performance without damaging the user experience, though it requires more time and access to ground truth data
- **More Rational Reliance:** Calibrating human self-confidence encourages people to follow the "higher confidence party" (either themselves or the AI) when they disagree.
- **Reduced Under-reliance:** Calibration significantly reduced **under-reliance** (rejecting correct AI advice), though it did not significantly impact over-reliance in this specific study.
- **Improved Team Performance:** Calibration led to higher initial and final accuracy, allowing the human-AI team to outperform the AI's standalone accuracy.

- The "C-C Matching" Analytical Framework

- The paper introduces a novel framework for diagnosing reliance issues by looking at **Confidence-Correctness (C-C) Matching**

Situation	Description	Resulting Behaviour
C-C Matched	High confidence + Correct OR Low confidence + Incorrect.	Likely leads to correct reliance.
C-C Mismatched	High confidence + Incorrect (Over-confident) OR Low confidence + Correct (Under-confident).	Frequently leads to incorrect reliance.

As Confidence Aligns: Understanding the Effect of AI Confidence on Human Self-confidence in Human-AI Decision Making (Li et al., 2025)

The authors demonstrate a phenomenon called confidence alignment: in human-AI collaboration, users' self-confidence tends to gravitate toward the confidence levels expressed by the AI. This alignment is problematic because it can occur without any actual change in the user's decision-making accuracy, leading to miscalibrated self-confidence.

- **Unidirectional Alignment:** During human-AI decision-making, participants' self-confidence levels tend to align with the AI's confidence levels.
- **Persistence:** This alignment is not just temporary; it persists in individual tasks even after the AI is no longer assisting the user.
- **Independence from Accuracy:** The alignment was not caused by users becoming more accurate (learning effects). Instead, it appears to be a form of social imitation where humans "copy" the AI's expressed certainty.
- **Paradigm Neutrality:** The degree of alignment was consistent regardless of the AI's role—whether it acted as an **Advisor**, a **Peer Collaborator**, or an **Automated Decision-maker** under human supervision.
- **The Power of Feedback:** Providing real-time correctness feedback **reduced** the degree of alignment. Feedback helps users anchor their confidence in their actual performance rather than just imitating the AI.
- **Worsened Calibration:** For most participants (Type B: overconfident but less confident than AI), alignment moved their confidence further away from their actual accuracy, degrading their calibration.
- **Improved Calibration:** For users who were underconfident, aligning with a more confident AI actually improved their calibration.
- **Consequences:** Poorer calibration due to alignment led to **inappropriate reliance** (e.g., ignoring correct AI advice or over-relying on wrong AI advice) and lowered the overall accuracy of the human-AI team.

The study analyzed three distinct ways humans and AI work together:

- **AI as Advisor:** AI gives suggestions; the human makes the final choice.
- **AI as Peer Collaborator:** Both provide decisions; a mechanism (like "highest confidence") picks the final one.
- **AI as Supervisor:** AI makes decisions automatically; the human observes and only intervenes if something goes wrong

Human Tutoring Improves the Impact of AI Tutor Use on Learning Outcomes (Gurung et al., 2025)

This study evaluates a year-long implementation of Human-AI Tutoring (a hybrid model) compared to AI-only Tutoring. The researchers used a quasi-experimental design to see if adding a "human in the loop" (via virtual Zoom sessions) could solve the scalability issues of traditional tutoring while outperforming standalone software.

The study compared 356 seventh-graders (Human-AI) against 317 students from the previous year (AI-only).

- **Grade Level Growth:** The Human-AI group was **0.36 grade levels ahead** of the AI-only group by the end of the year.
- **The "Time-on-Task" Multiplier:** This is the most critical finding. While the Human-AI group didn't automatically crush the state tests, they saw massive gains **if they actually used the software**.
    - For every **3.26 hours** (1 SD) spent in the AI tutor, Human-AI students gained **0.28 SD** on state tests.
    - AI-only students only gained **0.06 SD** for that same amount of time.
    - **Takeaway:** Human tutors don't just teach; they "unlock" the effectiveness of the AI software.

One of the biggest "wins" highlighted is the economic feasibility of this model compared to traditional "High-Impact Tutoring" (HIT).

Tutoring Model	**Estimated Annual Cost (per student)**
Traditional High-Impact Tutoring	$3,500 – $4,300
Human-AI Hybrid (this study)	**~$700**
AI-only (i-Ready)	~$30

The hybrid model uses a **1:4 tutor-to-student ratio** with undergraduate tutors rotating in a "round-robin" style. Even though each student only got **5–8 minutes** of direct human contact per session, it was enough to drive significant gains.

The data revealed a sobering trend: **Student engagement declined over the year** in both groups, but the drop-off was actually *more pronounced* in the Human-AI group.

- **The Theory:** The "novelty effect" wore off, or students learned how to "game" the system (disengage) while the tutor was busy in another Zoom breakout room.
- **The Lesson:** Human-AI tutoring isn't a "set it and forget it" solution. Tutors need better tools to see who is slacking off in real-time.

- **The "Floor" Effect:** Many students were 2+ years behind grade level. The researchers suggest the state tests (which target 7th-grade material) might have been too hard to accurately measure the progress of a student moving from a 4th-grade level to a 5th-grade level.
- **The Human Element:** The human tutors weren't just for math help; they were trained in **Social-Emotional Learning (SEL)** and relationship building. The study suggests the *motivational* support was just as important as the *academic* support.
- **The Metric of Success:** "Time-on-task" (how long the student stays in the program) is the best predictor of whether the tutoring will actually work.

Active Methods to possible try:

In the design of the study, we can compare up to 3 active strategies studying with AI, and have a passive control. Here are the selection of active strategies that we can choose from:

Socratic (WILL USE)

AI asks open-ended questions, where students justify their answer while connecting concepts. Students need to learn the direct facts (OpenAI's study mode).
ICAP framework (Chi & Wylie, 2014):

- Interactive → highest level, : dialogue & join construction of knowledge
- Constructive, student generate new ideas beyond learning materials (self explaining)
- active, → physical manip of focus attention taking notes, highlighting
- passive → lowest level : receive info without performing any action

Fakour & Imani 2025:

- Socratic questioning → method of asking thought provoking question than direct answers stimulating student reflection
- Critical Thinking Dimensions measured:
    - Analysis: break down complex infos
    - Evaluation: Assessing credibility / logical strength of argument
    - Inference: Logical conclusion from data
    - Self-Regulation: monitoring & correcting one's own cognitive process
- AI shouldn't be to replace humans but as a learning process fostering intellectual humility and persistent questioning.

Favero et al., 2024

- Bot uses fine tuning to ask structured thought provoking questions
- Uses smaller openSource models Llama-2 7B & 13B, address concerns on data privacy and democratized access to high quality education tools
- Ai finds when student is stuck and provide scaffolding → guides to student conclusion
- The bot can avoid repetition trap & is good at staying on topic
- Socratic method like this is good for theoretical concepts

In terms of for learning how to code, based on (*Alshaik et al., 2020*) socratic guides student to reason through the logic, key concepts of source code and how it functions, → students showed better improvement in programming concepts compared to their pre-test scores, breaks down complex code into logical segments that require cognitive processing

Reverse Tutoring (WILL USE)

Student is the teacher and explain concepts to the AI, the AI act like a 10 year old that asks clarifying questions ("Wait, but why does X happen")

- **Okita & Schwarts,** 2013 explore learning by teaching using a mechanism called Recursive Feedback, the tutor observe how their students apply what's been taught
- Protege effects → people learn harder when they know they must teach, significant learning gain occur after the teaching sessions → tutor observe the student's outcome

- Observing the students mistake/successes allows tutor to make inference about their own understanding, if student fails the tutor's own explanation may have been incomplete or flawed → triggering self correction
    - They do propose that having a 3rd party is important so that when teaching to a new person, the knowledge must be robust enough to be generalize

Learning by Teaching, Fioerlla et al., 2021:

- The act of preparing to teach triggers deep cognitive processes leading to better retention than simply studying
- 2 cognitive process:
    - Generative process → organize and integrate info in a way that makes sense
    - Metacognitive monitoring → identify gaps in your own knowledge when faced with a question or point that is difficult to explain
- Active explanation provides great benefit transgorming their understanding from surface level recognition to deep & flexible mastery

The role of teachable agents' personality traits on student-AI interactions and math learning (Lyu et al., 2025):

- Agents emphasizing openness (curiosity and preference for variety) leads to more student explanations and higher order cognitive engagement compared to extraversion or agree-ablness
- Extraversion-emphasis agents highly interactive but lead to shallower social interactions
- The most effective Ai tutor isn't a 1 size fis all, but have personality traits designed to trigger specific learning behaviors.

The Powerful Teacher: A Power Hypothesis for the Benefits of Learning-by-Teaching" (Wong & Lim, 2026):

- Role of teacher increases student sense of power which boosts cognitive engagement and learning outcome
- Group of explaining and teaching group showed better basic comprehension of material than notetaking group
- The authors conclude that the benefits of learning-by-teaching are not just cognitive (due to better organization of info) but also socio-psychological. By empowering students to see themselves as influential "teachers," educators can unlock higher levels of creativity and critical thinking that traditional study methods may not reach.

Scaffolder

AI gives problems to solve, student answers, if they get it wrong, AI dont give answer but hints, hint keeps getting larger the more the mistake

Shabani et al, 2010

Application of Vygotsky Zone of Proximal Development → distance between learner development level and their potential development (what they can do alone vs with guidance)

- Learning is a social process, traditional test only measure their past development, in contrast dynamic assessment interact within their ZPD to reveal future potential
- Instruction should be designed to encourage peer interaction so they can scaffold each other

Cognitive Conflict | Debate

Student explains a concept, AI debates and takes the incorrect stance, arguing with the student, the student must defend their knowledge & point out AI's mistakes
(Singer, 2018)
Study is using **nature of science**

- Combining cognitive conflict success is limited by how students perceive their owb=n errors, educators are encouraged to use targeted conceptual questions that force students to confront & resolve their internal logic.

(Limon, 2001(https://doi.org/10.1016/S0959-4752(00)00037-2))

- A bit difficult, it doesn't guarantee
- Student overlooks info that doesn't fit their mental model, might even twist new evidence to support their existing theory
- To make it work, student must have high metacognitive awareness, monitor own understanding, other than that student must also have:
    - Prior Knowledge, and motivation to engage in the debates

(Sumadji & Yuwono, 2020)

- 3 stages of resolution
    - Present a problem outside of the student logic
    - Make sure student recognize their answer is inconsistent
    - Guide the student to reorganize their cognitive structure

Developing cognitive conflict is highly effective, though complex, for math, but requires teachers to have deep understanding of common misconceptions.

Erroneous Example | Error-Driven Learning

AI gives summarization recaps that has deliberate errors, and student must find the mistakes and correct the AI's mistakes
Finding from the paper (Tsovaltzi et al., 2010)

- Benefits **vary by grade level,** higher grade gain cognitive and conceptually, lower grade showed metacognitive learning gains, while middle grade showed no significant benefit
- Must have the role of **scaffolding to assist in detecting and correcting errors** (learner must **have prior knowledge**)
- Self-explaining both correct & incorrect examples strengthen correct strategies & weakening incorrect ones
- Mathematics teachers remain skeptical of the practice (student might assimilate incorrect points)

ZONGS (TO BE EXCLUDED):

When to Ask a Question: Understanding Communication Strategies in Generative AI Tools (Park et al., 2025)
The authors conclude that the method of obtaining user information is just as essential as the information itself. Developers should not rely solely on the "chatbot" iterative model; instead, they should leverage distributional data to actively determine the optimal interaction strategy based on whether they are prioritizing efficiency, fairness, or a balance of both.